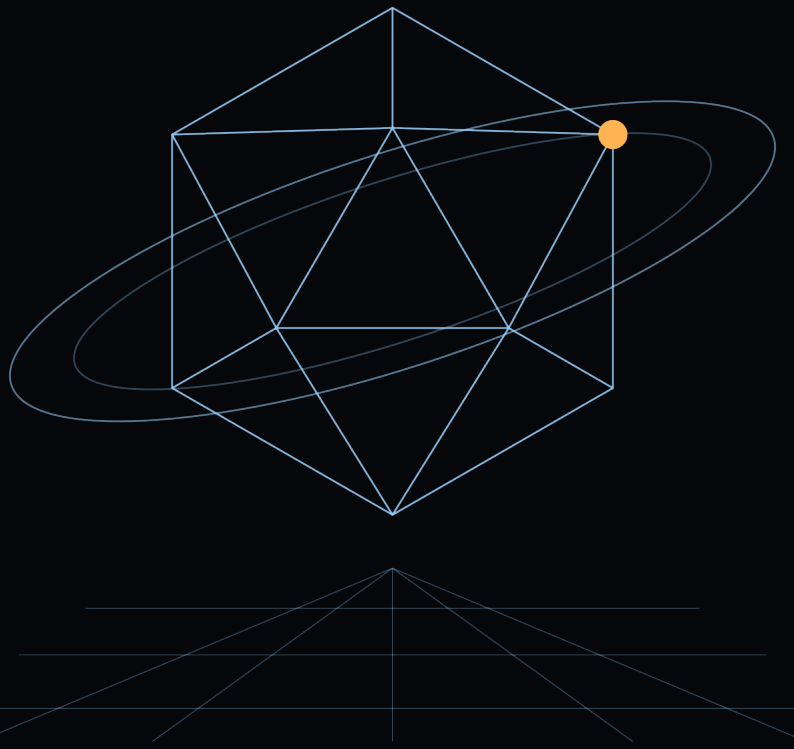
DECODING
DATA SCIENCE

CPD UK BOOTCAMP | DECOBOT | DATA UNDERSTANDING

Know the data before you build the bot



Structured and unstructured data, read by eye and read with code.

UNSTRUCTURED
DDS_Leave_Policy_v2.pdf

STRUCTURED
dds_leave.db

WHERE THIS FITS

The 3-day programme at a glance

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DAY 1

Control the AI

System prompt and clarity. No files.

DAY 2

Give it the policy

RAG, chunking. Uses DDS_Leave_Policy_v2.pdf.

DAY 3

Give it data and tools

SQL, three tools, agent. Uses the PDF and dds_leave.db.

BEFORE DAY 2 | THIS SESSION

Data understanding: read the PDF and the database yourself first. About 30 minutes.

> One story throughout: leave at Decoding Data Science, 100 employees, Dubai.

WHY START HERE

Understand the data first

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01

Wrong data, wrong answer

A chatbot is only as good as the files behind it. A missing page or a broken table becomes a confident wrong answer.

02

The right tool for the job

Rules live in documents. Numbers live in databases. Knowing which is which decides what you build.

03

Trust you can check

If you can show where an answer came from, people trust it. That starts with knowing what is in each file.

> A SMALL STORY
>
> The bot says 22 days. The policy says 22. The database says 2 left. Who is right? Only the person who read all of them.

THE TWO SOURCES

Two kinds of data, two jobs

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UNSTRUCTURED

1 policy PDF

Shape	Free text, pages, headings
Read with	Your eyes, or pypdf
Used for	RAG search, Day 2
Answers	What is the rule?
Example	How much notice does annual leave need?

STRUCTURED

dds_leave.db

Shape	Tables, rows, columns
Read with	A DB viewer, or sqlite3 and pandas
Used for	SQL tool, Day 3
Answers	What is the number?
Example	How many days does Aisha Khan have left?

Three ways to read data

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01 BY EYE

Read it yourself

Open the PDF and scroll. Open the database in a viewer and look at ten rows.

Good for: first impressions and spotting oddities.

Limit: slow, and it does not scale.

02 WITH CODE

Let Python read it

pypdf, sqlite3 and pandas read everything the same way every time.

Good for: counting, checking and repeating.

Limit: you must know what to ask.

03 BOTH, IN ORDER

Eye first, then code

What you notice by eye tells you what to check in code.

Good for: real understanding.

Limit: it takes time. Spend it.

RULE OF THUMB

Never build on data you have not read yourself.

Read the PDF

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BY EYE

Open each file

- Can you select and copy the text?
- Note the headings, tables and lists
- Check the version and the date
- Find the sections that matter for leave

WITH CODE

```
from pypdf import PdfReader
reader = PdfReader("DDS_Leave_Policy_v2.pdf")
pages = [p.extract_text() for p in reader.pages]
print(len(pages), "pages")
print(pages[0][:300])
```

KEEP THE PAGE

extract_text() returns one string per page. Keep the page number: it becomes the citation later.

Profile the documents

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01 PAGES

4 pages

A short file. Page 4 is the lightest with about 150 words.

02 WORDS

About 1,770 words

Roughly 11,000 characters, so about 12 chunks of 1,200 characters.

03 HEADINGS

12 numbered sections

From 1. Leave Entitlements to 12. Governance. Good places to cut.

04 KEYWORDS

leave, days, notice

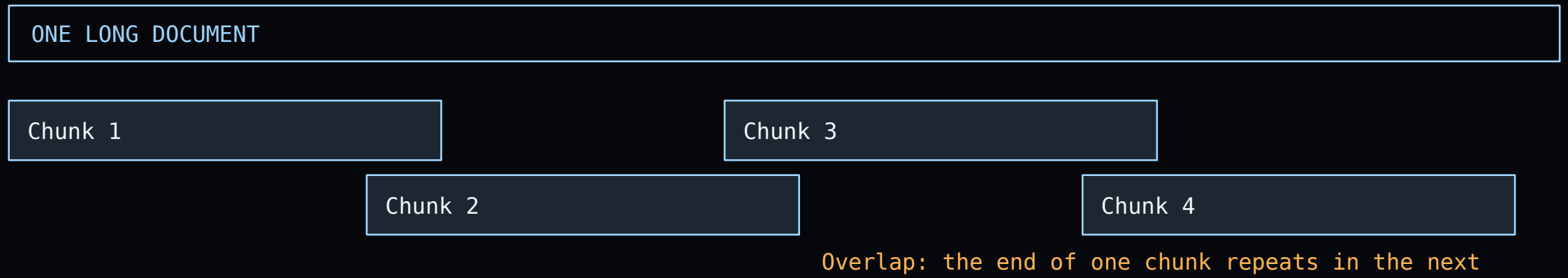
The top words match the story: leave, days, calendar, working, notice.

BEFORE YOU TRUST A PDF, CHECK

- Text extracts cleanly
- Headings are numbered
- Tables did not break
- Version: 2.0, 1 July 2026

Chunking: cut text to search it

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CHUNK SIZE

1,200 characters

Too big and the search gets noisy. Too small and a rule is cut in half.

OVERLAP

200 characters

A sentence on a boundary still appears whole in one chunk.

THE LESSON

Measure, do not guess

On Day 2 we test sizes with real questions. Here: 12 chunks at 1,200.

Read the database structure

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TABLE | 100 ROWS

employees

employee_id
name
department
job_title
branch
join_date
annual_total
annual_used
annual_booked
annual_remaining
sick_total
sick_used
sick_remaining

TABLE | 273 ROWS

leave_requests

request_id
employee_id (links to employees)
leave_type
start_date
end_date
working_days
status
requested_on

HOW BALANCES WORK

Stored, and computed

annual_used: approved, started by 28 Oct 2026
annual_booked: approved, starts later
annual_remaining: 22 minus used minus booked
sick_remaining: 15 minus used
Pending, Rejected, Cancelled do not count

HOW IT FITS

employee_id links requests to people. Balances in employees were computed from the requests, so the two tables agree.

Read the database with code

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BY EYE

Open it in a viewer

- Use a free tool such as DB Browser for SQLite
- Look at the first ten rows of each table
- Count the rows
- Read the column names out loud

WITH CODE

```
import sqlite3, pandas as pd
con = sqlite3.connect("dds_leave.db")
emp = pd.read_sql("SELECT * FROM employees", con)
reqs = pd.read_sql("SELECT * FROM leave_requests", con)
print(emp.shape, reqs.shape)
print(emp.annual_remaining.describe())
```

SAME DATA

Pandas turns each table into a data frame you can count, filter and chart.

Check quality before you trust it

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CHECK	ISSUES FOUND
Duplicate employee IDs	0
End date before start date	0
Requests for an unknown employee	0
Zero or negative working days	0
Remaining days that do not add up	0
Requested after 28 Oct 2026	14

WORTH A LOOK

Odd, not always wrong

14 requests are dated after today, and 9 approved requests overlap an earlier one for the same person. Synthetic data, so we note it.

Always ask: error, or how the data was made?

What the data says

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100
employees in 8 departments

273
leave requests in 2026

203
annual requests, 64 sick

14.5
average annual days left

PATTERN

Annual leave is about three quarters of all requests: 203 of 273.

SPREAD

Days left range from 0 to 22, so some people are close to running out.

DEMO CAST

Aisha Khan has 2 days left. John Doe has 17 and one pending request.

PUTTING BOTH TOGETHER

Which source answers which question

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QUESTION	SOURCE	WHY
How many days notice for annual leave?	**PDF (RAG)**	A rule in the policy
Who approves my leave?	**PDF (RAG)**	A rule in the policy
How many days does Aisha Khan have left?	**DATABASE (SQL)**	A personal number
Show John Doe’s leave requests	**DATABASE (SQL)**	Personal records
Can Aisha take 12 days?	**BOTH**	2 days left, plus the approval rules
What is the probation period?	**CHECK FIRST**	The policy may not cover it

PUTTING BOTH TOGETHER

Do the two sources agree?

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THE POLICY SAYS

22 and 15

working days: 22 annual leave and 15 sick leave a year (section 1).

THE DATABASE STORES

22 • 15

All 100 employees have annual_total 22 and sick_total 15.

> RESULT
>
> They agree. One catch in the policy itself: section 1 points to section 4.4 for extended illness, but that content is in 5.4. Bots should quote sections.

YOUR TURN

Practice and next steps

DECODING DATA SCIENCE

TRY IT IN THE NOTEBOOK

01 Set the chunk size to 400, then 2,500. How does the chunk count change?

02 Which employees took the most sick days? (Hint: 4 days.)

03 Write a SQL query: pending requests per department. (Answer: 4)

04 Does the policy define the probation period? Check first.

WHAT COMES NEXT

DAY 2

Chunk, embed and search the PDF with RAG

DAY 3

Wrap SQL queries as tools for the agent

NOTEBOOK

Day2_RAG_and_Chunking.ipynb in Google Colab